

Nikki Phach

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Skills

TypeScript, JavaScript, Python, SQL, React, React Router, Redux, HTML/CSS, XState, TailwindCSS, shadcn/ui, Responsive Web Design, FastAPI, Flask, REST APIs, Phaser, Vitest, Unit Testing, Zod, npm, Docker, Git/GitHub, Java, C, Linux/Unix, PyTorch, NumPy, Pandas, scikit-learn, HuggingFace, Jenkins, Machine Learning, Natural Language Processing, Computer Vision, Data Structures/Algorithms, Database Design, Data Processing

Work Experience

Konami Gaming Inc., Software Engineer I

April 2025 - present

- Eliminated ~20 hours of the math team's workload per week by developing an internal tool that automated maximum liability calculations, enabling product managers to directly request and update game liability data.
- Enabled promotional free spin capability for 60+ online casinos by developing a universal feature that integrates with a third-party content provider's API and UI wrapper.
- Ensured compliance with Brazil gaming regulations by introducing a user confirmation flow for automatic spins within gameplay.
- Improved flexibility of RTP (return-to-player) configuration by redesigning the loading flow to externalize metadata and adding ability to override the RTP display on game launch.
- Released new game title, All Aboard Panda Pennies, by cloning an existing game and extending it with responsive orientation support, full-screen background handling, and character group animations for compatible with its unique large-reel feature.
- Prepared entry into the German market by developing methodology for treating progressive pools as static jackpots, using scripts to manipulate progressive databases and implementing a customizable RTP display.
- Delivered 10+ staging site launches and 70+ game deployments by managing site-specific configurations across multiple jurisdictions.
- Shipped 15+ game re-releases, resolving 85+ issues to ensure production stability.

Projects

Kotoba Tag!

kotoba-tag.com

- Implemented an answer validation pipeline with ~95% accuracy by fine-tuning a SentenceTransformer model on 4M+ word pairs that were scraped, labeled and organized for cross-lingual definition similarity.
- Curated a 5,800+ word Japanese vocabulary bank with JLPT filtering, part-of-speech validation and syllable matching logic.
- Deployed a production application with XState, React Router, and a responsive shadcn/ui layout.
- Built a FastAPI backend on Render proxying HuggingFace inference and Jisho.org lookups, with cold-start warmup and retry handling.
- Enhanced gameplay with scoring multipliers, configurable game settings, and dictionary detail lookup.
- Maintained reliability with a Vitest suite covering syllable matching, definition validation and game flow.

Personal Website

nikki.ph

- Built and deployed a Tamagotchi-inspired portfolio SPA with React, TypeScript, Vite, Tailwind CSS, and GitHub Actions CI/CD to GitHub Pages.
- Orchestrated an LCD zoom transition in Motion with viewport-aware geometry, game-inspired inventory UX, and Canvas pixel trails.

Image Analysis: Skin Detection

Spring 2024

- Improved a skin detection pipeline from ~73% to ~98% accuracy by applying color-space analysis and dynamic thresholding, refactoring the core algorithm, and adding async image processing to reduce runtime.
- Built a ground-truth skin-mask test dataset with Photoshop and implemented a Metrics class to benchmark baseline performance and evaluate detection accuracy against labeled annotations.

Education

Boston University, M.Sc in Software Engineering for AI

part time | expected May 2028

University of Massachusetts Boston, B.Sc in Computer Science

May 2024